

A Scatter Plot with Linear Regression analysis is a quick way to visualize trends in continuous data variables.

Step 1: Select samples

Drag and drop terms of interest into the Subset boxes on the **Comparison** tab. Include all terms to be used in the analysis.

Click on the **Summary Statistics** tab to review the number of samples that will be included in the analysis. Confirm that the set of samples available is sufficient for running a meaningful analysis.

Step 2: Set up analysis

Click on the **Advanced Workflow** tab and select "Scatter Plot with Linear Regression" from the Analysis dropdown list.

Drag terms from the Navigate Concepts tree into the Independent Variable (1) and Dependent Variable (2) boxes. Both boxes should contain continuous variables (e.g. eGFR).

Click the "Run" button (3). A popup box will show a progress bar as the data is compiled and analyzed by tranSMART. Some analyses may take several minutes to run.

Comparison Summary Statistics Advanced Workflow Workspace SmartR

Analysis: Scatter Plot with Linear Regression

Cohorts:
Subset 1: INCLUDE ('Public Studies\JDRF_v1\Q2 Clinical data and measurements\Lab measurements\Metabolic panel\Urea Nitrogen (mmol/L)') AND INCLUDE ('Public Studies\JDRF_v1\Q2 Clinical data and measurements\Lab measurements\A1c (percent)')

Variable Selection

1 Independent Variable
Select a continuous variable from the Data Set Explorer Tree and drag it into the box.

123 A1c (percent) (24)

2 Dependent Variable
Select a continuous variable from the Data Set Explorer Tree and drag it into the box.

123 Urea Nitrogen (mmol/L) (24)

High Dimensional Data Clear

☐ Perform log₁₀ transformation on independent variable. (Do not apply to high dimensional data. Negative values will be ignored.)

3 Run

Step 3: Review results

Results will appear underneath the variable selection boxes (4). The Independent variable will appear along the X-axis and the Dependent variable will be the Y-axis.

Scroll below the box plot to see numerical Linear Regression results (5).

5 Linear Regression Result

Number of Subjects	24
Intercept	47.6
Slope	-32.6
r-squared	0.0316
adjusted r-squared	-0.0124
p-value	0.406

